

Verification Test Protocol 1

Graphic Equalizer Frequency Response Verification

Project	EE22005 Home Audio System (HAS) — Project Week 4
Group	Group HAS (BTA26, FS881, AC3847)
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Version	v1.0
Prepared for	Feasibility Study — Synthesis Evidence (Verification)

Version	Date	Author	Description of changes
v1.0	30/04/2026	Group HAS (BTA26, FS881, AC3847)	Initial issue of equalizer verification protocol — covers BPF1, BPF2, BPF3 and the summed full-EQ response.

Test ID	PW4_HAS_VER_001
Document purpose	Formal verification protocol for the three-band graphic equalizer covering individual bandpass filter response, full-EQ neutral and full-boost responses, and ripple measurement.
Related specification	S1 — Equalizer frequency response; S4 — Independent per-band control.
Verification type	Bench verification with function generator and oscilloscope.
Primary equipment	Function generator (RIGOL DG1022 or equivalent), four-channel oscilloscope, digital multimeter, oscilloscope cursors, breadboarded MFB equalizer with TL072 op-amps.
Prepared by	Group HAS (BTA26 — build lead; FS881 — design lead; AC3847 — evidence custodian)
Revision / Version	v1.0 — Initial issue, end of Day 2

1. Objective

To verify that the three Multiple Feedback (MFB) bandpass filters and the summing amplifier together meet target specification S1: each bandpass filter centred on its target frequency within $\pm 15\%$, providing up to 12 dB of gain with the band's gain control fully boosted, and presenting a flat 0 dB response across the audio band when all three controls are set to neutral. The protocol also verifies independent per-band control as required by S4 [1], [2].

2. Requirement traceability

Ref	Requirement summary	Acceptance criterion	Compliance basis
S1	BPF1 (bass) centre frequency	100 Hz $\pm 15\%$	Frequency-response sweep with function generator and oscilloscope.
S1	BPF2 (mid) centre frequency	1 kHz $\pm 15\%$	Frequency-response sweep with function generator and oscilloscope.
S1	BPF3 (treble) centre frequency	8 kHz $\pm 15\%$	Frequency-response sweep with function generator and oscilloscope.
S1	Maximum per-band gain at full boost	≥ 12 dB ± 1 dB	Cursor measurement at peak of each BPF response.
S1	Neutral-position ripple (all three pots centred)	< 4 dB across 100 Hz–10 kHz	Frequency-response sweep at neutral; cursor min/max.
S4	Independent per-band control	Adjacent bands change by < 2 dB at their centre frequencies when one band is moved cut $\rightarrow$ boost	Repeat measurement of two bands with the third swept end-to-end.

3. Roles and responsibilities

Test lead (FS881): owns measurement procedure and recording. Build lead (BTA26): operates equipment, makes any wiring corrections under change control. Evidence custodian (AC3847): captures oscilloscope screenshots and saves them to the QMS Quality_Control folder using the naming convention defined in [10]. Any deviation requires agreement of all three members and is logged in the change record before continuing.

4. Equipment and preconditions

The equalizer build must be powered by the bench supply at ± 9 V (or as specified by the design schematic).

The function generator output amplitude shall be set to 100 mV pp and held constant across all sweeps.

Oscilloscope channel coupling shall be AC for response measurements and DC for offset checks.

All three EQ potentiometers shall start in the centre (neutral) position before any sweep begins.

Pre-power continuity verification (Test Protocol 2 — Pre-Power) must already have been signed off.

5. Controlled variables

Function generator amplitude held constant at 100 mV pp.

Same op-amps (TL072) and same component reels used for all three filters.

Same probe and channel used between successive measurements where possible.

Bench supply rails confirmed within ± 5 % at the start of each measurement block.

6. Detailed procedure

6.1 Individual BPF response — BPF1 (100 Hz)

Disconnect BPF2 and BPF3 from the summing node. Set BPF1 gain control to full boost. Sweep the function generator from 20 Hz to 1 kHz in approximately 10 steps (20, 50, 80, 100, 150, 200, 300, 500, 800, 1000 Hz). At each step, record the input and output amplitudes from the oscilloscope. Calculate gain in dB. Identify the centre frequency as the point of maximum gain. Confirm the centre frequency is within 100 Hz ± 15 % and the peak gain is 12 dB ± 1 dB.

6.2 Individual BPF response — BPF2 (1 kHz)

As 6.1, but sweep from 200 Hz to 5 kHz. Confirm centre at 1 kHz ± 15 % and peak gain 12 dB ± 1 dB.

6.3 Individual BPF response — BPF3 (8 kHz)

As 6.1, but sweep from 2 kHz to 20 kHz. Confirm centre at 8 kHz ± 15 % and peak gain 12 dB ± 1 dB.

6.4 Full-EQ neutral response

Reconnect all three BPFs to the summing amplifier. Set all three potentiometers to centre. Sweep from 50 Hz to 20 kHz. The combined output amplitude shall not vary by more than 4 dB across the audio band — this verifies that the three filters sum cleanly with no peaks or nulls at their crossover points.

6.5 Full-EQ full-boost response

With all three BPFs reconnected, set all three potentiometers to full boost. Repeat the 50 Hz to 20 kHz sweep and confirm three distinct peaks at approximately 100 Hz, 1 kHz and 8 kHz, each at +12 dB ± 1 dB.

6.6 Independent-control check

Set BPF1 and BPF3 to neutral. Sweep BPF2 from full cut to full boost in three discrete steps. At each step, measure the gain at 100 Hz and at 8 kHz. The change in gain at these adjacent-band centre frequencies shall be less than 2 dB across the full BPF2 range.

6.7 Rework and re-verification

If any single test fails its acceptance criterion, isolate the affected BPF, verify the breadboard wiring against the schematic, replace any suspect components from E12 lab stock, and re-run the affected procedure section in full. Failures and rework actions shall be recorded in Section 7 (Data recording template).

7. Data recording template

Check item	Expected criterion	Measured value	Pass / Fail	Notes / rework
BPF1 centre frequency	100 Hz ± 15 %	_____ Hz	_____	
BPF1 peak gain	12 dB ± 1 dB	_____ dB	_____	
BPF2 centre frequency	1 kHz ± 15 %	_____ Hz	_____	
BPF2 peak gain	12 dB ± 1 dB	_____ dB	_____	
BPF3 centre frequency	8 kHz ± 15 %	_____ Hz	_____	
BPF3 peak gain	12 dB ± 1 dB	_____ dB	_____	
Full-EQ neutral ripple	< 4 dB	_____ dB	_____	
Full-EQ full-boost peaks	Three peaks +12 dB ± 1 dB	see screenshot	_____	
Independent-control check	Adjacent change < 2 dB	_____ dB	_____	

8. Acceptance criteria and decision rule

All BPF centre frequencies shall fall within ± 15 % of their nominal target.

Peak gain on each band shall be 12 dB ± 1 dB at full boost.

Neutral-position ripple shall be below 4 dB across the audio band.

Independent-control check shall show < 2 dB change at adjacent-band centres.

Any failed item requires correction and re-verification before the protocol is signed off.

9. Safety notes

No earphones shall be connected during this protocol — output is monitored on the oscilloscope only.

Bench supplies shall remain at ± 9 V; do not exceed the TL072 absolute maximum rails.

Function-generator amplitude must be confirmed at 100 mV pp before injection to avoid clipping the input stage.

10. Conclusion statement

The graphic equalizer is considered to have passed verification of S1 and S4 only when all checks in this protocol have been completed, recorded, and passed without unresolved anomalies.

Pass/fail evidence shall be recorded as oscilloscope screenshots saved per [10] and referenced in the Verification Results Summary.

References

- [1] International Electrotechnical Commission, Audio/video, information and communication technology equipment — Part 1: Safety requirements, IEC 62368-1:2023, 4th ed., May 2023.
- [2] H. Zumbahlen (Ed.), Linear Circuit Design Handbook. Burlington, MA, USA: Newnes / Analog Devices, 2008, ch. 8.
- [10] Department of Electronic and Electrical Engineering, EE22005 Engineering Practice and Design — Project Week 4 Brief, Univ. of Bath, 2026.